MAE 269: HW 2

Due April 17, 2026 at midnight

1 Representing inertia in different reference frames

The generalized inertia matrix for a rigid link in the body frame of the link can be expressed as

$$\mathcal{G}_b = \begin{bmatrix} \mathcal{I}_b & 0 \\ 0 & \mathrm{m}I \end{bmatrix}$$

and the total kinetic energy of motion of that link is related to the body twist and inertia matrix as

$$\frac{1}{2}\mathcal{V}_b^{\mathrm{T}}\mathcal{G}_b\mathcal{V}_b$$

1. Use an energy argument to demonstrate that the generalized inertia in the space frame $\mathcal{G}_s$ is related to $\mathcal{G}_b$ through the following formula

$$\mathcal{G}_s = \left[\mathrm{Ad}_{T_{\mathrm{bs}}}\right]^{\mathrm{T}} \mathcal{G}_{\mathrm{b}} \left[\mathrm{Ad}_{T_{\mathrm{bs}}}\right]$$

2. Expand the equation above into 2x2 block form expressing the components of $\mathcal{G}_s$ in terms of R, m, $\mathcal{I}_b$, and $[p]$.
3. A single link pendulum has length l, mass m, and body frame inertia $\mathcal{I}_b$. The body frame is located half way along the link. Compute the body and space twists for the center of mass location.
4. Demonstrate by hand that kinetic energy is the same when calculated in both body and space frames.

2 Simulating rigid body dynamics

Simulate the torque-free Euler equation in the body reference frame to solve for $\omega_B(t)$ as a function of time.

1. Write a helper function called `cross_matrix(vec)` which takes in a vector v and returns the cross-product matrix representation $[v]_\times$.
2. Write a helper function called `uncross_matrix(mat)` which takes in a cross-product matrix m and returns the vector components.
3. Write a dynamics function called `omega_dot(t, omega, I_B)` which is a function of time (unused), omega, and body-system moment of inertia I_B. The function should compute $\dot{\omega}$ from the torque-free Euler equation

$$I_B\dot{\omega}_B + [\omega_B]_\times I_B\omega_B = 0 \quad \rightarrow \quad \dot{\omega}_B = -I_B^{-1}\left([\omega_B]_\times I_B\omega_B\right)$$

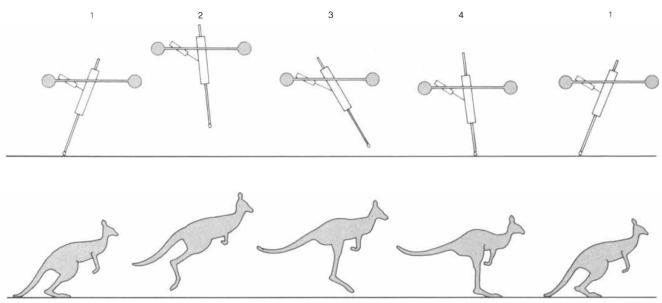
   and should return $\dot{\omega}_B$ as a row vector.
4. Use the sample code from lecture to numerically integrate the equation of motion ($\dot{\omega} = -I_B^{-1}\left([\omega]_\times I_B\omega\right)$) to solve for $\omega_B(t)$. Use a moment of inertia of

$$I_B = \begin{bmatrix} 10 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{1}$$

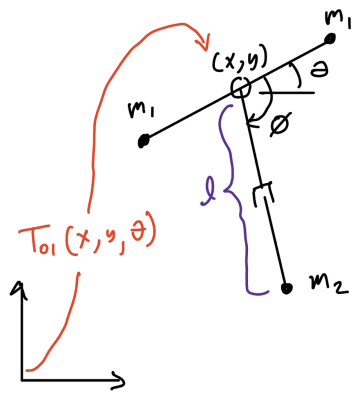
   Provide a plot of the components of ω_B versus time over a 10s time period, with $w_B(t=0) = [0.9., 0.1, 0]^T$ and $w_B(t=0) = [0.1., 0.9, 0]^T$.
5. Calculate the angular kinetic energy over time and provide a plot of energy versus time.

3 Euler-Lagrange equations for a simple hopper

The early hoppers of Marc Raibert were simple "two-link" bodies with an extendable leg. A dynamic model of this robot is shown below right, which consists of a body link with two masses m_1 that are a distance w from the center rotational joint. A second link is attached at the center rotational joint and this link can be changed in length, l, and has a mass m_2 located at the end.



HOPPER IN MOTION operates cyclically, as all legged systems do; the leg alternates between periods of support and periods of flight. At the left the machine is about to begin a leap. While it is in the air the leg swings forward at the hip in preparation for the next landing. At touchdown the leg spring shortens to its minimum length to provide for the next leap. A ground-contact sensor acts as a trigger for the vertical-control program. The machine also has feedback loops to control attitude and balance in synchrony with the vertical control. Like a pogo stick, the machine can balance only while it hops. The hop is like the kangaroo's movement Muybridge called a ricochet.



1. Write down the shape and pose variables that describe the full configuration of this system.
2. What are the dimensions of the mass matrix?
3. Write out the kinetic energy of the system in terms of the configuration variables.
4. Write out the potential energy of the system in terms of the configuration variables.
5. Using the sympy methods from class, derive the dynamics of this system using the Euler-Lagrange approach.
6. Provide the element of the mass matrix at row 2, column 3.
7. Using the methods from class write Python code to numerically integrate for the angles x, y, θ, ϕ, l with input torques $\tau_\phi = \sin(\omega t)$ and $\tau_l = \cos(\omega t)$. Provide a plot of ϕ over 5 cycles of oscillation.

4 Paper review

Please read the following paper and describe the methods used to model the system.

- **Required** Raibert, M. H. (1984). Hopping in legged systems — Modeling and simulation for the two-dimensional one-legged case. IEEE Transactions on Systems, Man, and Cybernetics, SMC-14(3), 451–463.